Vectors in Classical Physics

A Research-Level Tutorial

Based on

Mathematics of Classical and Quantum Physics
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Chapter 1: Expanded with Complete Proofs

Generated using physics-text-to-tutorial skill

January 21, 2026

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Chapter 1

Vectors in Classical Physics

B&F Ch. 1

1.1 The Scalar Product

B&F §1.3, pp. 5

1.1.1 Motivation and Context

The scalar (or "dot" or "inner") product is one of the most fundamental operations in vector analysis. It allows us to:

- Define the *length* of a vector without reference to coordinates
- Characterize *orthogonality* (perpendicularity) of vectors
- Project one vector onto another
- Connect geometric and algebraic descriptions of vectors

The scalar product will later generalize to the *inner product* on abstract vector spaces (Chapter 4), providing the foundation for Hilbert space and quantum mechanics.

1.1.2 Definition and Basic Properties

B&F p. 5

Definition 1.1.1 (Scalar Product — Geometric)**.** The scalar product of two vectors $\mathbf{x}$ and $\mathbf{y}$ is the real number

$$\mathbf{x} \cdot \mathbf{y} = |\mathbf{x}|\,|\mathbf{y}| \cos\theta, \tag{1.1.1}$$

where θ is the angle between the two vectors, measured from $\mathbf{x}$ to $\mathbf{y}$.

Intuition

The scalar product measures "how much" of $\mathbf{y}$ points in the direction of $\mathbf{x}$, scaled by the length of $\mathbf{x}$. Equivalently, it is $|\mathbf{x}|$ times the projection of $\mathbf{y}$ onto $\mathbf{x}$.

From the definition, we immediately obtain:

Proposition 1.1.1 (Basic Properties)**.** *The scalar product satisfies:*

(i) ***Commutativity:*** $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$

(ii) ***Relation to length:*** $|\mathbf{x}| = (\mathbf{x} \cdot \mathbf{x})^{1/2}$

(iii) ***Orthogonality criterion:*** $\mathbf{x} \cdot \mathbf{y} = 0$ *if and only if* $\mathbf{x} \perp \mathbf{y}$ *(or one is zero)*

Proof. **(i)** Since $\cos\theta$ is an even function and the angle from $\mathbf{x}$ to $\mathbf{y}$ equals the angle from $\mathbf{y}$ to $\mathbf{x}$:

$$\mathbf{x}\cdot\mathbf{y} = |\mathbf{x}||\mathbf{y}|\cos\theta = |\mathbf{y}||\mathbf{x}|\cos\theta = \mathbf{y}\cdot\mathbf{x}.$$

(ii) Setting $\mathbf{y} = \mathbf{x}$ in (1.1.1), we have $\theta = 0$, so $\cos\theta = 1$:

$$\mathbf{x}\cdot\mathbf{x} = |\mathbf{x}|^2.$$

(iii) If neither vector is zero, then $\mathbf{x}\cdot\mathbf{y} = 0$ implies $\cos\theta = 0$, hence $\theta = \pi/2$. ■

1.1.3 Distributivity: A Complete Geometric Proof

The following property is crucial but its proof is often omitted.

Theorem 1.1.2 (Distributivity of the Scalar Product)**.** *For any vectors* $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$*:*

$$\mathbf{x}\cdot(\mathbf{y}+\mathbf{z}) = \mathbf{x}\cdot\mathbf{y} + \mathbf{x}\cdot\mathbf{z}. \tag{1.1.2}$$

Key Step

Byron & Fuller note that "the reader will find it instructive to prove this by geometrical construction" *[Source: B&F p. 6]*. We now provide that proof in full.

Proof. We prove this using the projection interpretation of the scalar product.

Step 1: Projection interpretation. For any vector $\mathbf{v}$, the quantity $\mathbf{x}\cdot\mathbf{v}$ equals $|\mathbf{x}|$ times the signed length of the projection of $\mathbf{v}$ onto the direction of $\mathbf{x}$. Let $\hat{\mathbf{x}} = \mathbf{x}/|\mathbf{x}|$ be the unit vector in the direction of $\mathbf{x}$. Then:

$$\mathbf{x}\cdot\mathbf{v} = |\mathbf{x}|\cdot(\text{projection of } \mathbf{v} \text{ onto } \hat{\mathbf{x}}) = |\mathbf{x}|\cdot|\mathbf{v}|\cos\theta.$$

Step 2: Projection is a linear operation. Consider projecting onto a line L in the direction of $\mathbf{x}$. For vectors $\mathbf{y}$ and $\mathbf{z}$:

- Let $\mathbf{y}'$ be the projection of $\mathbf{y}$ onto L
- Let $\mathbf{z}'$ be the projection of $\mathbf{z}$ onto L
- Let $(\mathbf{y}+\mathbf{z})'$ be the projection of $\mathbf{y}+\mathbf{z}$ onto L

Step 3: Geometric construction. Place the tail of $\mathbf{z}$ at the head of $\mathbf{y}$ (this is vector addition). Drop perpendiculars from:

- The tail of $\mathbf{y}$ to L (point A)
- The head of $\mathbf{y}$ = tail of $\mathbf{z}$ to L (point B)
- The head of $\mathbf{z}$ to L (point C)

Then:

$$\begin{aligned}
\text{proj}_L(\mathbf{y}) &= \overrightarrow{AB} \\
\text{proj}_L(\mathbf{z}) &= \overrightarrow{BC} \\
\text{proj}_L(\mathbf{y}+\mathbf{z}) &= \overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}
\end{aligned}$$

Step 4: Conclusion. Since projection onto L is additive, and the scalar product is $|\mathbf{x}|$ times this projection:

$$\mathbf{x} \cdot (\mathbf{y} + \mathbf{z}) = |\mathbf{x}| \cdot \text{proj}_L(\mathbf{y} + \mathbf{z}) = |\mathbf{x}| \cdot [\text{proj}_L(\mathbf{y}) + \text{proj}_L(\mathbf{z})] = \mathbf{x} \cdot \mathbf{y} + \mathbf{x} \cdot \mathbf{z}.$$

■

1.1.4 Algebraic Form in Cartesian Coordinates

For Cartesian basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, orthonormality gives:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}, \tag{1.1.3}$$

where δ_{ij} is the **Kronecker delta**:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}. \tag{1.1.4}$$

B&F p. 7

Theorem 1.1.3 (Component Form of Scalar Product)**.** *For* $\mathbf{x} = x_i\mathbf{e}_i$ *and* $\mathbf{y} = y_j\mathbf{e}_j$ *(Einstein summation convention):*

$$\mathbf{x} \cdot \mathbf{y} = x_i y_i = \sum_{i=1}^{3} x_i y_i. \tag{1.1.5}$$

Proof. Using distributivity and the orthonormality of basis vectors:

$$\begin{aligned} \mathbf{x} \cdot \mathbf{y} &= (x_i\mathbf{e}_i) \cdot (y_j\mathbf{e}_j) \\ &= x_i y_j (\mathbf{e}_i \cdot \mathbf{e}_j) \\ &= x_i y_j \delta_{ij} \\ &= x_i y_i. \end{aligned} \tag{1.1.6}$$

In the last step, the Kronecker delta "kills" all terms except those where $i = j$. ■

1.2 Rotation of the Coordinate System: Orthogonal Transformations

B&F §1.4, pp. 8

1.2.1 Motivation

A fundamental principle of physics is that physical laws must not depend on the orientation of our coordinate system. This leads us to study how vector components change under rotation of coordinates, characterized by *orthogonal transformations*.

1.2.2 The Rotation Matrix

Consider two Cartesian bases $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$ with the same origin. Each primed basis vector can be expressed in terms of the unprimed:

$$\mathbf{e}'_i = a_{ij}\mathbf{e}_j, \quad (j = 1, 2, 3), \tag{1.2.1}$$

where

$$a_{ij} \equiv \mathbf{e}'_i \cdot \mathbf{e}_j \tag{1.2.2}$$

are the **direction cosines** (cosines of angles between basis vectors).

The nine numbers a_{ij} form the **rotation matrix**:

$$R \equiv (a_{ij}) = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}. \tag{1.2.3}$$

1.2.3 Orthogonality Relations: Complete Derivation

Theorem 1.2.1 (Orthogonality Relations)**.** *The elements of a rotation matrix satisfy:*

$$a_{ki}a_{kj} = \delta_{ij} \tag{1.2.4}$$
$$a_{ik}a_{jk} = \delta_{ij} \tag{1.2.5}$$

(summation over k implied).

Key Step

Byron & Fuller state that "it is left to the reader to show" equation (1.2.5) by expanding basis vectors *[Source: B&F p. 10]*. We provide the complete proof.

Proof. **Proof of (1.2.4):** Since the primed basis is orthonormal, $\mathbf{e}'_i \cdot \mathbf{e}'_j = \delta_{ij}$. Substituting (1.2.1):

$$\begin{aligned}\delta_{ij} &= \mathbf{e}'_i \cdot \mathbf{e}'_j \\ &= (a_{ik}\mathbf{e}_k) \cdot (a_{jl}\mathbf{e}_l) \\ &= a_{ik}a_{jl}(\mathbf{e}_k \cdot \mathbf{e}_l) \\ &= a_{ik}a_{jl}\delta_{kl} \\ &= a_{ik}a_{jk}.\end{aligned}$$

Proof of (1.2.5): We can also express unprimed basis vectors in terms of primed ones. Since (1.2.2) gives $a_{ij} = \mathbf{e}'_i \cdot \mathbf{e}_j$, we have:

$$\mathbf{e}_j = (\mathbf{e}_j \cdot \mathbf{e}'_i)\mathbf{e}'_i = a_{ij}\mathbf{e}'_i. \tag{1.2.6}$$

Now use orthonormality of unprimed basis, $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$:

$$\begin{aligned}\delta_{ij} &= \mathbf{e}_i \cdot \mathbf{e}_j \\ &= (a_{ki}\mathbf{e}'_k) \cdot (a_{lj}\mathbf{e}'_l) \\ &= a_{ki}a_{lj}(\mathbf{e}'_k \cdot \mathbf{e}'_l) \\ &= a_{ki}a_{lj}\delta_{kl} \\ &= a_{ki}a_{kj}.\end{aligned}$$

Relabeling indices ($k \to k$, and noting the symmetry), this is equivalent to $a_{ik}a_{jk} = \delta_{ij}$. ■

Remark 1.2.1. In matrix notation, (1.2.4) says $R^\mathsf{T}R = I$ and (1.2.5) says $RR^\mathsf{T} = I$. Thus $R^\mathsf{T} = R^{-1}$: the inverse of a rotation matrix is its transpose.

1.2.4 Counting Independent Parameters

Proposition 1.2.2. *In n dimensions, an orthogonal matrix has $\frac{1}{2}n(n-1)$ independent parameters.*

Key Step

Byron & Fuller state "as the reader can verify" that the orthogonality relations reduce the degrees of freedom *[Source: B&F p. 10]*. Here is the explicit count.

Proof. An $n \times n$ matrix has n^2 elements. The orthogonality condition $R^\mathsf{T} R = I$ gives constraints.

The matrix $R^\mathsf{T} R$ is symmetric (since $(R^\mathsf{T} R)^\mathsf{T} = R^\mathsf{T} R$), so it has $\frac{1}{2}n(n+1)$ independent entries (diagonal plus upper triangle).

Setting $R^\mathsf{T} R = I$ gives $\frac{1}{2}n(n+1)$ independent equations.

Therefore, the number of free parameters is:

$$n^2 - \frac{1}{2}n(n+1) = \frac{2n^2 - n^2 - n}{2} = \frac{n^2 - n}{2} = \frac{n(n-1)}{2}.$$

For $n = 2$: $\frac{2 \cdot 1}{2} = 1$ parameter (one rotation angle φ).

For $n = 3$: $\frac{3 \cdot 2}{2} = 3$ parameters (the three Euler angles φ, θ, ψ). ■

1.2.5 Transformation of Vector Components

B&F p. 11

Theorem 1.2.3 (Component Transformation Law)**.** *If a vector* $\mathbf{x}$ *has components* x_j *in basis* $\{\mathbf{e}_j\}$ *and components* x'_i *in basis* $\{\mathbf{e}'_i\}$*, then:*

$$x'_i = a_{ij} x_j \tag{1.2.7}$$

$$x_i = a_{ji} x'_j \tag{1.2.8}$$

Proof. **Forward transformation:** The vector $\mathbf{x}$ is the same geometric object in both bases:

$$\mathbf{x} = x_j \mathbf{e}_j = x'_i \mathbf{e}'_i.$$

Taking the scalar product with $\mathbf{e}'_i$:

$$x'_i = \mathbf{x} \cdot \mathbf{e}'_i = x_j (\mathbf{e}_j \cdot \mathbf{e}'_i) = x_j a_{ij} = a_{ij} x_j.$$

Inverse transformation: Multiply (1.2.7) by a_{ki} and sum over i:

$$a_{ki} x'_i = a_{ki} a_{ij} x_j = \delta_{kj} x_j = x_k.$$

Relabeling $k \to i$ and $i \to j$ gives (1.2.8). ■

1.2.6 Invariance of the Scalar Product

B&F p. 12

Theorem 1.2.4 (Invariance under Orthogonal Transformations)**.** *The scalar product is invariant under orthogonal transformations:*

$$\mathbf{x}' \cdot \mathbf{y}' = \mathbf{x} \cdot \mathbf{y}.$$

Intuition

Geometrically obvious: rotating the coordinate system doesn't change lengths or angles. But the algebraic proof is important for generalization.

Proof. Using component transformation and orthogonality:

$$\begin{aligned}
\mathbf{x}' \cdot \mathbf{y}' &= x'_i y'_i \\
&= (a_{ij} x_j)(a_{ik} y_k) \\
&= a_{ij} a_{ik} x_j y_k \\
&= \delta_{jk} x_j y_k \\
&= x_j y_j \\
&= \mathbf{x} \cdot \mathbf{y}.
\end{aligned}$$

■

Corollary 1.2.5. *Orthogonal transformations preserve:*

(i) *Vector lengths:* $|\mathbf{x}'| = |\mathbf{x}|$

(ii) *Angles between vectors*

(iii) *Orthogonality:* $\mathbf{x} \perp \mathbf{y} \Leftrightarrow \mathbf{x}' \perp \mathbf{y}'$

1.2.7 Key Results Summary

- The scalar product $\mathbf{x} \cdot \mathbf{y} = |\mathbf{x}||\mathbf{y}| \cos\theta = x_i y_i$ is commutative and distributive
- Orthonormal bases satisfy $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$
- Rotation matrices satisfy $R^\mathsf{T} R = R R^\mathsf{T} = I$ (orthogonality)
- In 3D, rotations have 3 independent parameters (Euler angles)
- Components transform as $x'_i = a_{ij} x_j$; the inverse uses a_{ji}
- The scalar product is invariant under orthogonal transformations

1.3 The Vector Product

, pp. 14–17

1.3.1 Overview and Motivation

While the scalar product takes two vectors and produces a scalar, physics often requires an operation that takes two vectors and produces another vector. The *vector product* (or *cross product*) fills this need. It arises naturally in mechanics (torque, angular momentum), electromagnetism (the Lorentz force, Poynting vector), and throughout mathematical physics.

Intuition

The vector product $\mathbf{x} \times \mathbf{y}$ captures the "oriented area" of the parallelogram spanned by $\mathbf{x}$ and $\mathbf{y}$. The magnitude equals the area, while the direction is perpendicular to the plane—encoding the orientation via the right-hand rule.

1.3.2 Geometric Definition

Definition 1.3.1 (Vector Product — Geometric)**.** The *vector product* of two vectors $\mathbf{x}$ and $\mathbf{y}$ is written $\mathbf{z} \equiv \mathbf{x} \times \mathbf{y}$. In geometrical language, we define the *magnitude* of $\mathbf{z}$ by

$$|\mathbf{z}| = |\mathbf{x} \times \mathbf{y}| = |\mathbf{x}||\mathbf{y}| \sin\theta, \tag{1.3.1}$$

where θ is the angle measured from $\mathbf{x}$ to $\mathbf{y}$ such that $\theta \leq \pi$.

The *direction* of $\mathbf{z}$ is perpendicular to the plane containing $\mathbf{x}$ and $\mathbf{y}$, determined by the right-hand rule: with fingers swinging from $\mathbf{x}$ to $\mathbf{y}$, the thumb points in the direction of $\mathbf{z}$.

From the geometric definition, several properties follow immediately:

Proposition 1.3.1 (Anticommutativity)**.** *The vector product is anticommutative:*

$$(\mathbf{x} \times \mathbf{y}) = -(\mathbf{y} \times \mathbf{x}). \tag{1.3.2}$$

Proof. When we reverse the order, we swing fingers from $\mathbf{y}$ to $\mathbf{x}$ instead of from $\mathbf{x}$ to $\mathbf{y}$. Since $0 \leq \theta \leq \pi$, swinging in the opposite direction means the thumb points "down" instead of "up." Formally, the angle measured from $\mathbf{y}$ to $\mathbf{x}$ is $-\theta$ (or equivalently $2\pi - \theta$), so the direction reverses while the magnitude $|\mathbf{x}||\mathbf{y}| \sin\theta$ remains unchanged. ■

Corollary 1.3.2. *For any vector* $\mathbf{x}$*:*

$$\mathbf{x} \times \mathbf{x} = \mathbf{0}. \tag{1.3.3}$$

Proof. This follows geometrically since $\sin 0 = 0$. Alternatively, from anticommutativity: $\mathbf{x} \times \mathbf{x} = -\mathbf{x} \times \mathbf{x}$, which implies $2(\mathbf{x} \times \mathbf{x}) = \mathbf{0}$. ■

1.3.3 The Levi-Civita Symbol

B&F p. 15

To express the vector product algebraically, we introduce the fundamental *Levi-Civita symbol* (also called the permutation symbol or alternating tensor).

B&F p. 15, Eq.

Definition 1.3.2 (Levi-Civita Symbol)**.** The Levi-Civita symbol ε_{ijk} is defined by:

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3), \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3), \\ 0 & \text{otherwise (i.e., if any two indices are equal).} \end{cases} \tag{1.3.4}$$

Explicitly, the nonzero values are:

$$\varepsilon_{123} = \varepsilon_{231} = \varepsilon_{312} = +1 \quad \text{(cyclic permutations)}, \tag{1.3.5}$$

$$\varepsilon_{321} = \varepsilon_{213} = \varepsilon_{132} = -1 \quad \text{(anticyclic permutations)}. \tag{1.3.6}$$

The symmetry properties follow immediately from the definition:

$$\varepsilon_{ijk} = \varepsilon_{kij} = \varepsilon_{jki} = -\varepsilon_{jik} = -\varepsilon_{kji} = -\varepsilon_{ikj}. \tag{1.3.7}$$

1.3.4 The Fundamental Epsilon-Delta Identity

The following identity is one of the most important in vector and tensor analysis:

B&F p. 15, Eq.

Theorem 1.3.3 (Epsilon-Delta Identity)**.** *The product of two Levi-Civita symbols with one index contracted satisfies:*

$$\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}. \tag{1.3.8}$$

Proof requested p. 15: "We lea ification to the r

Proof. We prove this by systematic case analysis. First, note that both sides are antisymmetric under $i \leftrightarrow j$ and under $l \leftrightarrow m$, and both vanish unless (i,j) and (l,m) are distinct pairs.

Step 1: Establish when the LHS is nonzero.

For $\varepsilon_{ijk}\varepsilon_{klm}$ to be nonzero, we need:

- (i,j,k) to be a permutation of $(1,2,3)$, i.e., i,j,k all distinct
- (k,l,m) to be a permutation of $(1,2,3)$, i.e., k,l,m all distinct

Since both share the index k, and $\{i,j,k\} = \{l,m,k\} = \{1,2,3\}$, we must have $\{i,j\} = \{l,m\}$ as sets.

This means either $(i,j) = (l,m)$ or $(i,j) = (m,l)$.

Step 2: Case $(i,j) = (l,m)$.

When $i = l$ and $j = m$, we sum over $k \in \{1,2,3\} \setminus \{i,j\}$:

$$\sum_k \varepsilon_{ijk}\varepsilon_{klm} = \sum_k \varepsilon_{ijk}\varepsilon_{kij}.$$

For the unique k that makes both symbols nonzero (i.e., $k \neq i, k \neq j$):

$$\varepsilon_{ijk}\varepsilon_{kij} = \varepsilon_{ijk} \cdot \varepsilon_{ijk} = (\pm 1)^2 = 1.$$

The RHS gives: $\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl} = 1 \cdot 1 - 0 \cdot 0 = 1$. ✓

Step 3: Case $(i, j) = (m, l)$.

When $i = m$ and $j = l$:

$$\sum_k \varepsilon_{ijk}\varepsilon_{klm} = \sum_k \varepsilon_{ijk}\varepsilon_{kji}.$$

Using $\varepsilon_{kji} = -\varepsilon_{kij}$ (odd permutation):

$$\varepsilon_{ijk}\varepsilon_{kji} = \varepsilon_{ijk}(-\varepsilon_{kij}) = -(\varepsilon_{ijk})^2 = -1.$$

The RHS gives: $\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl} = 0 \cdot 0 - 1 \cdot 1 = -1$. ✓

Step 4: All other cases.

If $\{i, j\} \neq \{l, m\}$, then either some indices coincide making $\varepsilon_{ijk} = 0$ or $\varepsilon_{klm} = 0$, or there is no valid k. The LHS vanishes.

For the RHS: if $\{i, j\} \cap \{l, m\} = \emptyset$, then $\delta_{il} = \delta_{jm} = \delta_{im} = \delta_{jl} = 0$, so RHS $= 0$. ✓

If exactly one of $(i = l), (i = m), (j = l), (j = m)$ holds, then exactly one term in the RHS is nonzero, but the LHS vanishes because no valid k exists. Let's verify: suppose $i = l$ but $j \neq m$. Then RHS $= \delta_{jm} - 0 = 0$ since $j \neq m$. ✓

All cases agree, completing the proof. ■

Remark 1.3.1. The identity (1.3.8) can be remembered as a "determinant":

$$\varepsilon_{ijk}\varepsilon_{klm} = \begin{vmatrix} \delta_{il} & \delta_{im} \\ \delta_{jl} & \delta_{jm} \end{vmatrix}.$$

A useful corollary arises from contracting further indices:

Corollary 1.3.4 (Contractions of the Epsilon-Delta Identity)**.**

$$\varepsilon_{ijk}\varepsilon_{ilm} = \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}, \tag{1.3.9}$$

$$\varepsilon_{ijk}\varepsilon_{ijm} = 2\delta_{km}, \tag{1.3.10}$$

$$\varepsilon_{ijk}\varepsilon_{ijk} = 6. \tag{1.3.11}$$

Proof. For (1.3.9): This is just a relabeling of (1.3.8).

For (1.3.10): Set $l = j$ in (1.3.9):

$$\varepsilon_{ijk}\varepsilon_{ijm} = \delta_{jj}\delta_{km} - \delta_{jm}\delta_{kj} = 3\delta_{km} - \delta_{km} = 2\delta_{km}.$$

For (1.3.11): Set $m = k$ in (1.3.10):

$$\varepsilon_{ijk}\varepsilon_{ijk} = 2\delta_{kk} = 2 \cdot 3 = 6.$$

Alternatively: there are exactly 6 nonzero components of ε_{ijk}, each squaring to 1. ■

15–16

1.3.5 Algebraic Definition of the Vector Product

For a right-handed orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, the vector product of basis vectors is:

$$\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk}\mathbf{e}_k, \tag{1.3.12}$$

5, Eq. (1.32)

where summation over k is implied.

Definition 1.3.3 (Vector Product — Algebraic)**.** Given vectors $\mathbf{x} = x_j\mathbf{e}_j$ and $\mathbf{y} = y_k\mathbf{e}_k$, their vector product is:

$$\mathbf{z} = \mathbf{x} \times \mathbf{y} = x_j y_k \mathbf{e}_j \times \mathbf{e}_k = x_j y_k \varepsilon_{jki}\mathbf{e}_i = \varepsilon_{ijk} x_j y_k \mathbf{e}_i. \tag{1.3.13}$$

The ith component is:

$$z_i = (\mathbf{x} \times \mathbf{y})_i = \varepsilon_{ijk} x_j y_k. \tag{1.3.14}$$

Written out explicitly:

$$z_1 = x_2 y_3 - x_3 y_2, \tag{1.3.15}$$
$$z_2 = x_3 y_1 - x_1 y_3, \tag{1.3.16}$$
$$z_3 = x_1 y_2 - x_2 y_1. \tag{1.3.17}$$

This is often memorized via the formal determinant:

$$\mathbf{x} \times \mathbf{y} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}. \tag{1.3.18}$$

1.3.6 The BAC-CAB Rule and Vector Identities

The epsilon-delta identity makes proving vector identities almost mechanical. B&F p. 16, Eq.

Theorem 1.3.5 (BAC-CAB Rule)**.** *For any vectors* $\mathbf{x}, \mathbf{y}, \mathbf{z}$*:*

$$\mathbf{x} \times (\mathbf{y} \times \mathbf{z}) = \mathbf{y}(\mathbf{x} \cdot \mathbf{z}) - \mathbf{z}(\mathbf{x} \cdot \mathbf{y}). \tag{1.3.19}$$

B&F p. 16, Exa

Proof. We compute the ith component of the LHS:

$$[\mathbf{x} \times (\mathbf{y} \times \mathbf{z})]_i = \varepsilon_{ijk} x_j (\mathbf{y} \times \mathbf{z})_k \tag{1.3.20}$$
$$= \varepsilon_{ijk} x_j \varepsilon_{klm} y_l z_m \tag{1.3.21}$$
$$= \varepsilon_{ijk}\varepsilon_{klm} x_j y_l z_m. \tag{1.3.22}$$

Applying the epsilon-delta identity (1.3.8):

$$= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) x_j y_l z_m \tag{1.3.23}$$
$$= \delta_{il}\delta_{jm} x_j y_l z_m - \delta_{im}\delta_{jl} x_j y_l z_m \tag{1.3.24}$$
$$= y_i(x_j z_j) - z_i(x_j y_j) \tag{1.3.25}$$
$$= y_i(\mathbf{x} \cdot \mathbf{z}) - z_i(\mathbf{x} \cdot \mathbf{y}) \tag{1.3.26}$$
$$= [\mathbf{y}(\mathbf{x} \cdot \mathbf{z}) - \mathbf{z}(\mathbf{x} \cdot \mathbf{y})]_i. \tag{1.3.27}$$

Since this holds for each component i, the vector identity is proved. ■

Intuition

The mnemonic "BAC-CAB" comes from reading the formula as $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$. Note the pattern: the "outside" vector $\mathbf{A}$ dots with the other two.

6, Eq. (1.35)

1.3.7 The Scalar Triple Product

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Theorem 1.3.6 (Cyclic Property of Scalar Triple Product)**.** *For any vectors* $\mathbf{x}, \mathbf{y}, \mathbf{z}$*:*

$$\mathbf{x} \cdot (\mathbf{y} \times \mathbf{z}) = \mathbf{z} \cdot (\mathbf{x} \times \mathbf{y}) = \mathbf{y} \cdot (\mathbf{z} \times \mathbf{x}). \tag{1.3.28}$$

Proof. **Method 1: Direct computation using indices.**

The scalar triple product in index notation is:

$$\mathbf{x} \cdot (\mathbf{y} \times \mathbf{z}) = x_i(\mathbf{y} \times \mathbf{z})_i = x_i \varepsilon_{ijk} y_j z_k = \varepsilon_{ijk} x_i y_j z_k.$$

Similarly:

$$\mathbf{y} \cdot (\mathbf{z} \times \mathbf{x}) = \varepsilon_{ijk} y_i z_j x_k.$$

Relabeling dummy indices $(i, j, k) \to (k, i, j)$:

$$\varepsilon_{ijk} y_i z_j x_k = \varepsilon_{kij} y_k z_i x_j.$$

Since $\varepsilon_{kij} = \varepsilon_{ijk}$ (cyclic permutation preserves sign):

$$= \varepsilon_{ijk} x_j y_k z_i = \varepsilon_{ijk} x_i y_j z_k.$$

(The last equality uses that summed indices are dummies.)

For the third expression:

$$\mathbf{z} \cdot (\mathbf{x} \times \mathbf{y}) = \varepsilon_{ijk} z_i x_j y_k.$$

Relabeling $(i, j, k) \to (j, k, i)$:

$$\varepsilon_{ijk} z_i x_j y_k = \varepsilon_{jki} z_j x_k y_i = \varepsilon_{ijk} x_k y_i z_j = \varepsilon_{ijk} x_i y_j z_k. \quad \checkmark$$

Method 2: Determinant interpretation.

The scalar triple product equals the determinant:

$$\mathbf{x} \cdot (\mathbf{y} \times \mathbf{z}) = \begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix}.$$

A cyclic permutation of rows in a 3×3 determinant requires two row swaps (e.g., $R_1 \to R_2$, then $R_2 \to R_3$), each changing the sign. Two sign changes preserve the determinant's value:

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} = \begin{vmatrix} y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \\ x_1 & x_2 & x_3 \end{vmatrix} = \begin{vmatrix} z_1 & z_2 & z_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}.$$

These are precisely $\mathbf{x} \cdot (\mathbf{y} \times \mathbf{z})$, $\mathbf{y} \cdot (\mathbf{z} \times \mathbf{x})$, and $\mathbf{z} \cdot (\mathbf{x} \times \mathbf{y})$ respectively. ■

Physical Interpretation

The scalar triple product $\mathbf{x} \cdot (\mathbf{y} \times \mathbf{z})$ gives the (signed) volume of the parallelepiped with edges $\mathbf{x}, \mathbf{y}, \mathbf{z}$. The cyclic property reflects that the volume is independent of which edge we call "base" and which we call "height."

1.3.8 Transformation Properties: Vectors vs. Pseudovectors

B&F pp. 16–17

A crucial question is: does the vector product of two vectors transform like a vector under orthogonal transformations?

B&F p. 16

Theorem 1.3.7 (Vector Product Transforms as a Vector)**.** *If* $\mathbf{x}$ *and* $\mathbf{y}$ *transform as vectors under orthogonal transformations* ($x'_i = a_{ij}x_j$, $y'_k = a_{km}y_m$)*, then* $\mathbf{z} = \mathbf{x} \times \mathbf{y}$ *also transforms as a vector:*

$$z'_i = a_{ij}z_j. \tag{1.3.29}$$

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Proof. The key is the identity:

$$\varepsilon_{ijk}a_{1k} = \varepsilon_{lmn}a_{mi}a_{nj}, \tag{1.3.30}$$

which states how the Levi-Civita symbol "transforms."

Key Step

Identity (1.3.30) follows from the fact that $\det(A) = \varepsilon_{lmn}a_{1l}a_{2m}a_{3n}$, combined with $\det(A) = 1$ for proper rotations. Specifically, for any orthogonal matrix with $\det(A) = 1$:

$$\varepsilon_{ijk} = \varepsilon_{lmn}a_{li}a_{mj}a_{nk}.$$

Multiplying both sides by a_{1k} and using $a_{nk}a_{1k} = \delta_{n1}$:

$$\varepsilon_{ijk}a_{1k} = \varepsilon_{lmn}a_{li}a_{mj}\delta_{n1} = \varepsilon_{lm1}a_{li}a_{mj} = \varepsilon_{lmn}a_{mi}a_{nj}$$

(relabeling in the last step).

Now we compute z'_i:

$$\begin{aligned} z'_i &= \varepsilon_{ijk}x'_j y'_k && (1.3.31)\\ &= \varepsilon_{ijk}(a_{jn}x_n)(a_{km}y_m) && (1.3.32)\\ &= \varepsilon_{ijk}a_{jn}a_{km}x_n y_m. && (1.3.33) \end{aligned}$$

Using (1.3.30) (with appropriate index relabeling):

$$\begin{aligned} z'_i &= \varepsilon_{nml}a_{il}x_n y_m && (1.3.34)\\ &= a_{il}\varepsilon_{nml}x_n y_m && (1.3.35)\\ &= a_{il}(\mathbf{x} \times \mathbf{y})_l && (1.3.36)\\ &= a_{il}z_l = a_{ij}z_j. && (1.3.37) \end{aligned}$$

This is precisely the transformation law for a vector. ■

Warning

The above proof assumes $\det(A) = +1$ (proper rotations). Under *improper* orthogonal transformations (those with $\det(A) = -1$, such as reflections or inversions), the vector

product transforms differently!

Definition 1.3.4 (Polar and Axial Vectors).

- A *polar vector* (or *ordinary vector*) changes sign under inversion ($\mathbf{e}_i \to -\mathbf{e}_i$). Examples: position $\mathbf{r}$, momentum $\mathbf{p}$, force $\mathbf{F}$, electric field $\mathbf{E}$.
- An *axial vector* (or *pseudovector*) does *not* change sign under inversion. The vector product of two polar vectors is an axial vector. Examples: angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, torque $\mathbf{N} = \mathbf{r} \times \mathbf{F}$, magnetic field $\mathbf{B}$.

Verification that $\mathbf{x} \times \mathbf{y}$ *is a pseudovector.* Under inversion, each component of a polar vector changes sign: $x_i \to -x_i$ and $y_j \to -y_j$.

The components of $\mathbf{z} = \mathbf{x} \times \mathbf{y}$:

$$z_i = \varepsilon_{ijk} x_j y_k \to \varepsilon_{ijk}(-x_j)(-y_k) = \varepsilon_{ijk} x_j y_k = z_i.$$

Thus z_i does *not* change sign—the two minus signs cancel. This is the defining property of a pseudovector. ■

Physical Interpretation

The distinction between polar and axial vectors has physical significance. In the equation

$$\frac{d\mathbf{L}}{dt} = \mathbf{r} \times \mathbf{F} = \mathbf{N},$$

both sides are pseudovectors. Angular momentum and torque relate two pseudovectors, while $\mathbf{r}$ and $\mathbf{F}$ are polar vectors. This ensures the equation is consistent under coordinate inversions (parity transformations)—a requirement for physically meaningful laws.

Remark 1.3.2. The scalar product of a polar vector and an axial vector is a *pseudoscalar*—it changes sign under inversion. For example, if $\mathbf{a}$ is polar and $\mathbf{b}$ is axial, then $\mathbf{a} \cdot \mathbf{b} \to (-\mathbf{a}) \cdot \mathbf{b} = -\mathbf{a} \cdot \mathbf{b}$.

1.3.9 Key Results Summary

- The **vector product** $\mathbf{x} \times \mathbf{y}$ is perpendicular to both $\mathbf{x}$ and $\mathbf{y}$, with magnitude $|\mathbf{x}||\mathbf{y}| \sin\theta$
- The **Levi-Civita symbol** ε_{ijk} encodes the sign of permutations and enables algebraic manipulation
- The **epsilon-delta identity** $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ is essential for proving vector identities
- The **BAC-CAB rule**: $\mathbf{x} \times (\mathbf{y} \times \mathbf{z}) = \mathbf{y}(\mathbf{x} \cdot \mathbf{z}) - \mathbf{z}(\mathbf{x} \cdot \mathbf{y})$
- The **scalar triple product** is invariant under cyclic permutations
- The vector product of two polar vectors is a **pseudovector** (axial vector), which doesn't change sign under inversion

, pp. 17–19

1.4 A Vector Treatment of Classical Orbit Theory

1.4.1 Overview and Motivation

This section demonstrates the power of vector methods by deriving the classical orbit equation for a particle in an inverse-square central force. Rather than solving the differential equations of motion directly (as done in standard mechanics texts), we employ vector identities to obtain the orbit with remarkable elegance.

Intuition

The key insight is that in addition to energy and angular momentum, the inverse-square force has a *third* conserved vector quantity (the Laplace–Runge–Lenz vector). This "hidden" symmetry leads directly to the conic section orbits.

1.4.2 Conservation of Angular Momentum

B&F p. 17

Theorem 1.4.1 (Kepler's Second Law)**.** *Angular momentum is constant for motion in any central force field.*

Proof. The angular momentum is defined as

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\dot{\mathbf{r}}. \tag{1.4.1}$$

Differentiating with respect to time:

$$\dot{\mathbf{L}} = \dot{\mathbf{r}} \times \mathbf{p} + \mathbf{r} \times \dot{\mathbf{p}}. \tag{1.4.2}$$

The first term vanishes because $\dot{\mathbf{r}}$ and $\mathbf{p} = m\dot{\mathbf{r}}$ are parallel:

$$\dot{\mathbf{r}} \times \mathbf{p} = \dot{\mathbf{r}} \times m\dot{\mathbf{r}} = \mathbf{0}.$$

For the second term, Newton's second law gives $\dot{\mathbf{p}} = \mathbf{F}$. For a central force, $\mathbf{F} = f(r)\mathbf{n}$ where $\mathbf{n} = \mathbf{r}/r$ is the radial unit vector. Thus:

$$\mathbf{r} \times \dot{\mathbf{p}} = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times f(r)\mathbf{n} = f(r)(\mathbf{r} \times \mathbf{n}) = f(r)\left(\mathbf{r} \times \frac{\mathbf{r}}{r}\right) = \mathbf{0}.$$

Therefore $\dot{\mathbf{L}} = \mathbf{0}$, so $\mathbf{L}$ is constant. ■

Physical Interpretation

Since $\mathbf{L}$ is constant and $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is perpendicular to both $\mathbf{r}$ and $\mathbf{p}$, the motion is confined to a fixed plane (the plane perpendicular to $\mathbf{L}$). This is the mathematical content of Kepler's second law (equal areas in equal times).

1.4.3 The Inverse-Square Force

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We now specialize to the inverse-square central force:

$$m\frac{d\mathbf{v}}{dt} = -\frac{k}{r^2}\mathbf{n}, \tag{1.4.3}$$

where:

- $\mathbf{n} = \mathbf{r}/r$ is the unit vector in the radial direction
- $k > 0$ for attractive force (gravity: $k = Gm_1m_2$; electrostatics: $k = q_1q_2$ in cgs units for like charges)

Notation 1.4.1. We write the velocity as $\mathbf{v} = \dot{\mathbf{r}}$. In polar-like coordinates within the orbital plane:

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r}\mathbf{n} + r\dot{\mathbf{n}}. \tag{1.4.4}$$

The term $\dot{r}\mathbf{n}$ is the radial component; $r\dot{\mathbf{n}}$ is the tangential component (since $\mathbf{n}$ rotates as the particle moves).

1.4.4 Expression for Angular Momentum

Lemma 1.4.2. *The angular momentum can be written as:*

$$\mathbf{L} = mr^2(\mathbf{n} \times \dot{\mathbf{n}}). \tag{1.4.5}$$

Proof. Using (1.4.4):

$$\begin{aligned}
\mathbf{L} &= \mathbf{r} \times m\mathbf{v} && (1.4.6)\\
&= r\mathbf{n} \times m(\dot{r}\mathbf{n} + r\dot{\mathbf{n}}) && (1.4.7)\\
&= mr\dot{r}(\mathbf{n} \times \mathbf{n}) + mr^2(\mathbf{n} \times \dot{\mathbf{n}}) && (1.4.8)\\
&= \mathbf{0} + mr^2(\mathbf{n} \times \dot{\mathbf{n}}) && (1.4.9)\\
&= mr^2(\mathbf{n} \times \dot{\mathbf{n}}). && (1.4.10)
\end{aligned}$$

■

7–18, Eq. (1.40)

1.4.5 A Key Derivative

The following computation is the heart of the derivation:

Lemma 1.4.3.

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{L}) = k\dot{\mathbf{n}}. \tag{1.4.11}$$

Proof. Using the product rule:

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{L}) = \dot{\mathbf{v}} \times \mathbf{L} + \mathbf{v} \times \dot{\mathbf{L}}. \tag{1.4.12}$$

Since $\mathbf{L}$ is constant, $\dot{\mathbf{L}} = \mathbf{0}$, so:

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{L}) = \dot{\mathbf{v}} \times \mathbf{L}. \tag{1.4.13}$$

From the equation of motion (1.4.3):

$$\dot{\mathbf{v}} = -\frac{k}{mr^2}\mathbf{n}. \tag{1.4.14}$$

Thus:

$$\begin{aligned}
\dot{\mathbf{v}} \times \mathbf{L} &= -\frac{k}{mr^2}\mathbf{n} \times \mathbf{L} && (1.4.15)\\
&= -\frac{k}{mr^2}\mathbf{n} \times \left[mr^2(\mathbf{n} \times \dot{\mathbf{n}})\right] \quad \text{using (1.4.5)} && (1.4.16)\\
&= -k\,\mathbf{n} \times (\mathbf{n} \times \dot{\mathbf{n}}). && (1.4.17)
\end{aligned}$$

Applying the BAC-CAB rule (1.3.19):

$$\mathbf{n} \times (\mathbf{n} \times \dot{\mathbf{n}}) = \mathbf{n}(\mathbf{n} \cdot \dot{\mathbf{n}}) - \dot{\mathbf{n}}(\mathbf{n} \cdot \mathbf{n}). \tag{1.4.18}$$

Now, since $|\mathbf{n}| = 1$, we have $\mathbf{n} \cdot \mathbf{n} = 1$.

Also, differentiating $\mathbf{n} \cdot \mathbf{n} = 1$:

$$\frac{d}{dt}(\mathbf{n} \cdot \mathbf{n}) = 2\mathbf{n} \cdot \dot{\mathbf{n}} = 0, \tag{1.4.19}$$

so $\mathbf{n} \cdot \dot{\mathbf{n}} = 0$ (the derivative of a unit vector is perpendicular to it).

Therefore:

$$\mathbf{n} \times (\mathbf{n} \times \dot{\mathbf{n}}) = \mathbf{n} \cdot 0 - \dot{\mathbf{n}} \cdot 1 = -\dot{\mathbf{n}}. \tag{1.4.20}$$

Substituting back:

$$\dot{\mathbf{v}} \times \mathbf{L} = -k(-\dot{\mathbf{n}}) = k\dot{\mathbf{n}}. \tag{1.4.21}$$

■

1.4.6 The Laplace–Runge–Lenz Vector

B&F p. 18, Eq.

Integrating (1.4.11):

$$\mathbf{v} \times \mathbf{L} = k\mathbf{n} + \mathbf{C}, \tag{1.4.22}$$

where $\mathbf{C}$ is a constant vector (the constant of integration).

Definition 1.4.1 (Laplace–Runge–Lenz Vector)**.** The constant vector $\mathbf{C}$ appearing in (1.4.22) is called the *Laplace–Runge–Lenz vector* (or simply the LRL vector). It lies in the orbital plane and points from the focus toward the perihelion.

Physical Interpretation

The existence of this additional conserved vector is special to the $1/r^2$ force law. It is responsible for the fact that orbits are closed (Bertrand's theorem). In quantum mechanics, this symmetry explains the "accidental" degeneracy of the hydrogen atom energy levels.

1.4.7 Derivation of the Orbit Equation

B&F pp. 18–19

To find the orbit, we compute L^2:

$$L^2 = \mathbf{L} \cdot (\mathbf{r} \times m\mathbf{v}) = m\,\mathbf{r} \cdot (\mathbf{v} \times \mathbf{L}), \tag{1.4.23}$$

where we used the cyclic property of the scalar triple product (1.3.28).

Substituting (1.4.22):

$$\begin{aligned} L^2 &= m\,\mathbf{r} \cdot (k\mathbf{n} + \mathbf{C}) && (1.4.24)\\ &= mk\,\mathbf{r} \cdot \mathbf{n} + m\,\mathbf{r} \cdot \mathbf{C} && (1.4.25)\\ &= mk\,\mathbf{r} \cdot \frac{\mathbf{r}}{r} + m\,r\,\mathbf{n} \cdot \mathbf{C} && (1.4.26)\\ &= mkr + mr\,|\mathbf{C}|\cos\theta, && (1.4.27) \end{aligned}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$ (measured from the direction of $\mathbf{C}$).

Solving for r:

$$L^2 = mr(k + |\mathbf{C}|\cos\theta), \tag{1.4.28}$$

$$r = \frac{L^2/mk}{1 + (|\mathbf{C}|/k)\cos\theta} = \frac{A}{1 + \varepsilon\cos\theta}, \tag{1.4.29}$$

where we define:

$$A = \frac{L^2}{mk} \quad \text{(semi-latus rectum)}, \tag{1.4.30}$$

$$\varepsilon = \frac{|\mathbf{C}|}{k} \quad \text{(eccentricity)}. \tag{1.4.31}$$

8, Eq. (1.42)

Theorem 1.4.4 (Orbit Equation)**.** *The orbit of a particle in an attractive inverse-square central force is a conic section with one focus at the force center:*

$$r = \frac{A}{1 + \varepsilon\cos\theta}. \tag{1.4.32}$$

The type of conic depends on the eccentricity:

- $\varepsilon = 0$*: circle*
- $0 < \varepsilon < 1$*: ellipse*
- $\varepsilon = 1$*: parabola*
- $\varepsilon > 1$*: hyperbola*

18–19

1.4.8 Eccentricity in Terms of Energy

The magnitude of $\mathbf{C}$ can be related to the total energy E.

First, we compute $|\mathbf{v} \times \mathbf{L}|^2$ at perihelion (closest approach), where $\theta = 0$ and $\mathbf{v}$ is perpendicular to $\mathbf{r}$:

$$|\mathbf{v} \times \mathbf{L}|^2 = v^2 L^2. \tag{1.4.33}$$

At perihelion, $\mathbf{v} \perp \mathbf{L}$ (since $\mathbf{L}$ is perpendicular to the orbital plane). Now $|\mathbf{v} \times \mathbf{L}|^2 = v^2 L^2$ because the velocity is perpendicular to $\mathbf{L}$.

Using (1.4.22) at any point:

$$|\mathbf{v} \times \mathbf{L}|^2 = |k\mathbf{n} + \mathbf{C}|^2 = k^2 + 2k|\mathbf{C}|\cos\theta + |\mathbf{C}|^2. \tag{1.4.34}$$

Since $|\mathbf{v} \times \mathbf{L}| = vL$ (the velocity is always perpendicular to $\mathbf{L}$ by definition of angular momentum):

$$v^2 L^2 = k^2 + 2k\mathbf{n} \cdot \mathbf{C} + C^2. \tag{1.4.35}$$

The total energy is:

$$E = \frac{1}{2}mv^2 - \frac{k}{r}, \tag{1.4.36}$$

so $v^2 = \frac{2}{m}\left(E + \frac{k}{r}\right)$.

After considerable algebra (computing at perihelion where $r = A/(1+\varepsilon)$), one obtains:

$$\varepsilon = \frac{|\mathbf{C}|}{k} = \sqrt{1 + \frac{2L^2 E}{mk^2}}. \tag{1.4.37}$$

Corollary 1.4.5 (Orbit Type from Energy)**.**

- $E < 0$*:* $\varepsilon < 1$ *(ellipse, bound orbit)*

- $E = 0$: $\varepsilon = 1$ *(parabola, marginally unbound)*
- $E > 0$: $\varepsilon > 1$ *(hyperbola, unbound)*

Proof. From (1.4.37):

- If $E < 0$: the argument of the square root is less than 1, so $\varepsilon < 1$.
- If $E = 0$: $\varepsilon = 1$.
- If $E > 0$: the argument exceeds 1, so $\varepsilon > 1$.

The case $E = -mk^2/(2L^2)$ gives $\varepsilon = 0$, a circular orbit. ■

1.4.9 Key Results Summary

- **Angular momentum conservation**: $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is constant for any central force, confining motion to a plane
- **Laplace–Runge–Lenz vector**: For the $1/r^2$ force, $\mathbf{v} \times \mathbf{L} - k\mathbf{n} = \mathbf{C}$ is an additional conserved quantity
- **Conic section orbits**: $r = A/(1 + \varepsilon \cos\theta)$ describes ellipses, parabolas, or hyperbolas depending on eccentricity
- **Eccentricity–energy relation**: $\varepsilon = \sqrt{1 + 2L^2E/(mk^2)}$ connects orbit shape to energy
- **Vector methods**: The entire derivation used only the BAC-CAB rule, scalar triple product, and the constancy of $\mathbf{L}$—no differential equations were solved directly

1.5 Differential Operations on Scalar and Vector Fields

B&F §1.7, pp. 1

1.5.1 Overview and Motivation

This section introduces the fundamental differential operators of vector calculus: the gradient, divergence, curl, and Laplacian. These operators are essential tools in classical physics, appearing throughout electromagnetism, fluid dynamics, and potential theory.

Intuition

The key question for each operator is: *what does it measure?*

- **Gradient**: How fast and in what direction does a scalar field change?
- **Divergence**: Is there a source or sink of a vector field at a point?
- **Curl**: Is there rotation or circulation in a vector field?
- **Laplacian**: How does a value at a point differ from its neighborhood average?

1.5.2 Orthogonal Curvilinear Coordinates

B&F pp. 19–20,

Before defining the operators, we establish the coordinate framework.

Definition 1.5.1 (Orthogonal Curvilinear Coordinates)**.** A set of coordinates (q_1, q_2, q_3) is *orthogonal curvilinear* if:

1. The coordinate surfaces $q_i = \text{const}$ are mutually perpendicular
2. The line element takes the form
$$ds^2 = h_1^2\, dq_1^2 + h_2^2\, dq_2^2 + h_3^2\, dq_3^2, \tag{1.5.1}$$
where the *scale factors* h_i may depend on position.

Notation 1.5.1. The displacement along coordinate q_i alone is:

$$ds_i = h_i\, dq_i \quad \text{(no summation)}. \tag{1.5.2}$$

e 1.1, p. 20

Example 1.5.1 (Common Coordinate Systems)**.**

System	(q_1, q_2, q_3)	(h_1, h_2, h_3)
Cartesian	(x, y, z)	$(1, 1, 1)$
Cylindrical	(r, θ, z)	$(1, r, 1)$
Spherical	(r, θ, φ)	$(1, r, r\sin\theta)$

20–22

1.5.3 The Gradient

Definition 1.5.2 (Gradient)**.** The gradient of a scalar field φ is defined by the requirement:

$$\operatorname{grad}\varphi \cdot d\mathbf{s} = d\varphi, \tag{1.5.3}$$

where $d\varphi$ is the differential change in φ corresponding to an arbitrary displacement $d\mathbf{s}$.

Physical Interpretation

This definition says: the gradient is the vector such that its projection onto any direction gives the rate of change of φ in that direction.

From the definition, we derive:

$$d\varphi = |\operatorname{grad}\varphi|\,|d\mathbf{s}|\cos\theta, \tag{1.5.4}$$

1

where θ is the angle between $\operatorname{grad}\varphi$ and $d\mathbf{s}$.

Theorem 1.5.1 (Properties of the Gradient)**.**

1. *The gradient points in the direction of maximum rate of change of φ*
2. *Its magnitude equals the maximum rate of change: $|\operatorname{grad}\varphi| = |d\varphi/ds|_{\max}$*
3. *The gradient is perpendicular to level surfaces $\varphi = const$*

Proof. From (1.5.4):

1. $d\varphi$ is maximized when $\cos\theta = 1$, i.e., when $d\mathbf{s}$ is parallel to $\operatorname{grad}\varphi$.
2. At maximum, $d\varphi = |\operatorname{grad}\varphi|\,|d\mathbf{s}|$, so $|d\varphi/ds|_{\max} = |\operatorname{grad}\varphi|$.
3. On a level surface, $d\varphi = 0$ for displacements within the surface, so $\operatorname{grad}\varphi \perp d\mathbf{s}$ for all such displacements.

■

(1.50), p. 21

Theorem 1.5.2 (Gradient in Curvilinear Coordinates)**.** *The ith component of the gradient is:*

$$(\operatorname{grad}\varphi)_i = \frac{1}{h_i}\frac{\partial\varphi}{\partial q_i}. \tag{1.5.5}$$

Proof. From (1.5.3) with displacement only in the q_i direction:

$$(\operatorname{grad}\varphi)_i\, ds_i = d\varphi = \frac{\partial\varphi}{\partial q_i}\, dq_i.$$

Using $ds_i = h_i\, dq_i$ from (1.5.2):

$$(\operatorname{grad}\varphi)_i \cdot h_i\, dq_i = \frac{\partial\varphi}{\partial q_i}\, dq_i.$$

Dividing by $h_i\, dq_i$:

$$(\operatorname{grad}\varphi)_i = \frac{1}{h_i}\frac{\partial\varphi}{\partial q_i}.$$

■

B&F Eq. (1.53)

Notation 1.5.2 (The Del Operator)**.** In Cartesian coordinates, we define the symbolic vector operator:

$$\nabla_{\text{Cartesian}} \equiv \mathbf{e}_i\partial_i = \mathbf{e}_1\frac{\partial}{\partial x} + \mathbf{e}_2\frac{\partial}{\partial y} + \mathbf{e}_3\frac{\partial}{\partial z}. \tag{1.5.6}$$

Then $\operatorname{grad}\varphi = \nabla\varphi$.

Warning

The ∇ symbol is often used in all coordinate systems, but its component form differs from (1.5.6) in curvilinear coordinates. In general:

$$\operatorname{grad}\varphi = \nabla\varphi = \frac{1}{h_1}(\partial_1\varphi)\mathbf{e}_1 + \frac{1}{h_2}(\partial_2\varphi)\mathbf{e}_2 + \frac{1}{h_3}(\partial_3\varphi)\mathbf{e}_3.$$

1.5.4 The Divergence

B&F pp. 22–24

B&F Eq. (1.56)

Definition 1.5.3 (Divergence)**.** The divergence of a vector field $\mathbf{V}$ at a point is defined as:

$$\operatorname{div}\mathbf{V} = \lim_{\Delta\tau\to 0}\frac{1}{\Delta\tau}\oint_\sigma \mathbf{V}\cdot d\boldsymbol{\sigma}, \tag{1.5.7}$$

where $\Delta\tau$ is a volume enclosing the point, σ is its bounding surface, and $d\boldsymbol{\sigma} = \mathbf{n}\, d\sigma$ with $\mathbf{n}$ the outward unit normal.

Physical Interpretation

The divergence measures the *flux per unit volume* emerging from an infinitesimal region, or equivalently, the *source strength per unit volume* of the vector field.

B&F Eq. (1.57)

Theorem 1.5.3 (Gauss's Divergence Theorem)**.** *For any volume τ bounded by surface σ:*

$$\int_\tau div\,\mathbf{V}\, d\tau = \oint_\sigma \mathbf{V}\cdot d\boldsymbol{\sigma}. \tag{1.5.8}$$

Proof. Subdivide the volume into many small subregions $\Delta\tau_i$. For each:

$$\operatorname{div}\mathbf{V}\cdot\Delta\tau_i \approx \oint_{\sigma_i}\mathbf{V}\cdot d\boldsymbol{\sigma}.$$

Summing over all subregions:

$$\sum_i \operatorname{div}\mathbf{V}\cdot\Delta\tau_i \approx \sum_i \oint_{\sigma_i}\mathbf{V}\cdot d\boldsymbol{\sigma}.$$

On the left, this becomes $\int_\tau \operatorname{div}\mathbf{V}\, d\tau$ as $\Delta\tau_i \to 0$.

On the right, each internal surface element $d\boldsymbol{\sigma}$ appears twice (belonging to two adjacent subregions) with opposite orientations of $\mathbf{n}$. These contributions cancel, leaving only the integral over the external boundary σ. ■

B&F Eq. (1.58)

Example 1.5.2 (Continuity Equation)**.** Let ρ be fluid density and $\mathbf{v}$ the velocity field. Conservation of mass requires:

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) = 0. \tag{1.5.9}$$

(1.59), p. 24 This says: the rate of density decrease equals the divergence of the mass flux.

Theorem 1.5.4 (Divergence in Curvilinear Coordinates)**.**

$$div\,\mathbf{V} = \frac{1}{h_1 h_2 h_3}\left[\frac{\partial(h_2 h_3 V_1)}{\partial q_1} + \frac{\partial(h_3 h_1 V_2)}{\partial q_2} + \frac{\partial(h_1 h_2 V_3)}{\partial q_3}\right]. \tag{1.5.10}$$

23–24

Proof. Consider an infinitesimal volume bounded by coordinate surfaces at $q_i = a_i$ and $q_i = a_i + \delta q_i$ for $i = 1, 2, 3$.

The volume element is $d\tau = h_1 h_2 h_3\, \delta q_1\, \delta q_2\, \delta q_3$.

For the flux through the face at $q_1 = a_1 + \delta q_1$ (normal in $+q_1$ direction):

$$\int_{a_2}^{a_2+\delta q_2} dq_2 \int_{a_3}^{a_3+\delta q_3} dq_3\, h_2 h_3\, V_1(a_1 + \delta q_1, q_2, q_3).$$

For the face at $q_1 = a_1$ (normal in $-q_1$ direction):

$$-\int_{a_2}^{a_2+\delta q_2} dq_2 \int_{a_3}^{a_3+\delta q_3} dq_3\, h_2 h_3\, V_1(a_1, q_2, q_3).$$

Combining and expanding to first order in δq_1:

$$\int dq_2\, dq_3 \left[\frac{\partial(h_2 h_3 V_1)}{\partial q_1}\right]_{q_1=a_1} \delta q_1.$$

As $\delta q_2, \delta q_3 \to 0$, this becomes:

$$\left[\frac{\partial(h_2 h_3 V_1)}{\partial q_1}\right] \delta q_1\, \delta q_2\, \delta q_3.$$

Adding the contributions from all three pairs of faces and dividing by $d\tau = h_1 h_2 h_3\, \delta q_1\, \delta q_2\, \delta q_3$ yields (1.5.10). ■

(1.60), p. 24

Corollary 1.5.5 (Divergence in Cartesian Coordinates)**.** *With $h_1 = h_2 = h_3 = 1$:*

$$div\,\mathbf{V} = \partial_i V_i = \frac{\partial V_1}{\partial x} + \frac{\partial V_2}{\partial y} + \frac{\partial V_3}{\partial z} = \nabla_{Cartesian} \cdot \mathbf{V}. \tag{1.5.11}$$

24–27

1.5.5 The Curl

(1.62), p. 24

Definition 1.5.4 (Curl)**.** The component of $\operatorname{curl}\mathbf{V}$ normal to a surface element is:

$$(\operatorname{curl}\mathbf{V}) \cdot \mathbf{n} = \lim_{\Delta\sigma \to 0} \frac{1}{\Delta\sigma} \oint_{\lambda} \mathbf{V} \cdot d\boldsymbol{\lambda}, \tag{1.5.12}$$

where $\Delta\sigma$ is the area bounded by the closed curve λ, and $\mathbf{n}$ is determined by the right-hand rule applied to the direction of integration around λ.

Physical Interpretation

The curl measures the *circulation per unit area*, or the tendency of the vector field to induce rotation. A small paddle wheel placed in a velocity field will rotate with angular velocity proportional to the curl.

B&F Eq. (1.63)

Theorem 1.5.6 (Stokes's Theorem)**.** *For any surface σ bounded by curve λ:*

$$\int_\sigma (curl\,\mathbf{V}) \cdot d\boldsymbol{\sigma} = \oint_\lambda \mathbf{V} \cdot d\boldsymbol{\lambda}. \tag{1.5.13}$$

Proof. Subdivide the surface into many small elements $\Delta\sigma_i$. For each:

$$(\mathrm{curl}\,\mathbf{V}) \cdot \mathbf{n}_i\,\Delta\sigma_i \approx \oint_{\lambda_i} \mathbf{V} \cdot d\boldsymbol{\lambda}.$$

Summing over all elements: on the left we get $\int_\sigma (\mathrm{curl}\,\mathbf{V}) \cdot d\boldsymbol{\sigma}$.

On the right, each internal edge appears twice with opposite orientations, so contributions cancel. Only the external boundary λ remains. ■

B&F Example 1

Example 1.5.3 (Rotating Fluid)**.** For a fluid rotating with constant angular velocity $\boldsymbol{\omega}$ about a vertical axis, the velocity is $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = \omega r\,\hat{\boldsymbol{\theta}}$ in cylindrical coordinates.

Using the curl definition with a circular path of radius r:

$$\mathbf{k} \cdot \mathrm{curl}\,\mathbf{v} = \lim_{r\to 0} \frac{1}{\pi r^2} \oint \omega r\,\hat{\boldsymbol{\theta}} \cdot r\,d\theta\,\hat{\boldsymbol{\theta}} = \lim_{r\to 0} \frac{\omega r \cdot 2\pi r}{\pi r^2} = 2\omega.$$

The curl is $2\boldsymbol{\omega}$—twice the angular velocity.

B&F Eq. (1.67)

Theorem 1.5.7 (Curl in Curvilinear Coordinates)**.**

$$curl\,\mathbf{V} = \frac{1}{h_2 h_3}[\partial_2(h_3 V_3) - \partial_3(h_2 V_2)]\mathbf{e}_1 + \textit{cyclic permutations}. \tag{1.5.14}$$

More explicitly:

$$(curl\,\mathbf{V})_1 = \frac{1}{h_2 h_3}\left[\frac{\partial(h_3 V_3)}{\partial q_2} - \frac{\partial(h_2 V_2)}{\partial q_3}\right], \tag{1.5.15}$$

$$(curl\,\mathbf{V})_2 = \frac{1}{h_3 h_1}\left[\frac{\partial(h_1 V_1)}{\partial q_3} - \frac{\partial(h_3 V_3)}{\partial q_1}\right], \tag{1.5.16}$$

$$(curl\,\mathbf{V})_3 = \frac{1}{h_1 h_2}\left[\frac{\partial(h_2 V_2)}{\partial q_1} - \frac{\partial(h_1 V_1)}{\partial q_2}\right]. \tag{1.5.17}$$

B&F Eq. (1.64)–(1.67)

Proof. We compute $(\mathrm{curl}\,\mathbf{V})_1$ using (1.5.12) with a rectangular loop in the q_2q_3-plane.

Consider an infinitesimal rectangle with corners at (q_2, q_3), $(q_2 + dq_2, q_3)$, $(q_2 + dq_2, q_3 + dq_3)$, $(q_2, q_3 + dq_3)$.

The line integral has four contributions:

$$\begin{aligned}
\text{Bottom } (q_3 = \text{const}): &\quad V_2\,h_2\,dq_2 \\
\text{Right } (q_2 + dq_2 = \text{const}): &\quad V_3(q_2 + dq_2)\,h_3(q_2 + dq_2)\,dq_3 \\
\text{Top } (q_3 + dq_3 = \text{const}): &\quad -V_2(q_3 + dq_3)\,h_2(q_3 + dq_3)\,dq_2 \\
\text{Left } (q_2 = \text{const}): &\quad -V_3\,h_3\,dq_3
\end{aligned}$$

Expanding to first order and combining:

$$\oint \mathbf{V} \cdot d\boldsymbol{\lambda} = \left[\frac{\partial(h_3 V_3)}{\partial q_2} - \frac{\partial(h_2 V_2)}{\partial q_3}\right] dq_2\, dq_3.$$

The area element is $d\sigma = h_2 h_3\, dq_2\, dq_3$, so:

$$(\operatorname{curl} \mathbf{V})_1 = \frac{1}{h_2 h_3}\left[\frac{\partial(h_3 V_3)}{\partial q_2} - \frac{\partial(h_2 V_2)}{\partial q_3}\right].$$

The other components follow by cyclic permutation. ■

(1.65)–(1.66),

Corollary 1.5.8 (Curl in Cartesian Coordinates)**.** *With $h_i = 1$:*

$$(curl\,\mathbf{V})_i = \varepsilon_{ijk}\partial_j V_k, \tag{1.5.18}$$

or equivalently:

$$curl\,\mathbf{V} = \nabla_{Cartesian} \times \mathbf{V} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \partial_1 & \partial_2 & \partial_3 \\ V_1 & V_2 & V_3 \end{vmatrix}. \tag{1.5.19}$$

28–30

1.5.6 The Laplacian

(1.68), p. 28

Definition 1.5.5 (Laplacian of a Scalar Field)**.** The Laplacian is defined as the divergence of the gradient:

$$\text{Laplacian}\,\varphi \equiv \operatorname{div}(\operatorname{grad}\varphi) \equiv \nabla \cdot (\nabla\varphi) \equiv \nabla^2\varphi. \tag{1.5.20}$$

(1.68), p. 28

Theorem 1.5.9 (Laplacian in Curvilinear Coordinates)**.**

$$\nabla^2\varphi = \frac{1}{h_1 h_2 h_3}\left[\partial_1\left(\frac{h_2 h_3}{h_1}\partial_1\varphi\right) + \partial_2\left(\frac{h_3 h_1}{h_2}\partial_2\varphi\right) + \partial_3\left(\frac{h_1 h_2}{h_3}\partial_3\varphi\right)\right]. \tag{1.5.21}$$

Proof. From (1.5.5), the gradient has components $(\operatorname{grad}\varphi)_i = \frac{1}{h_i}\partial_i\varphi$.

Applying (1.5.10) with $V_i = \frac{1}{h_i}\partial_i\varphi$:

$$\begin{aligned}\nabla^2\varphi &= \frac{1}{h_1 h_2 h_3}\left[\partial_1\left(h_2 h_3 \cdot \frac{1}{h_1}\partial_1\varphi\right) + \partial_2\left(h_3 h_1 \cdot \frac{1}{h_2}\partial_2\varphi\right) + \partial_3\left(h_1 h_2 \cdot \frac{1}{h_3}\partial_3\varphi\right)\right] \\ &= \frac{1}{h_1 h_2 h_3}\left[\partial_1\left(\frac{h_2 h_3}{h_1}\partial_1\varphi\right) + \partial_2\left(\frac{h_3 h_1}{h_2}\partial_2\varphi\right) + \partial_3\left(\frac{h_1 h_2}{h_3}\partial_3\varphi\right)\right].\end{aligned}$$

■

(1.69), p. 28

Corollary 1.5.10 (Laplacian in Cartesian Coordinates)**.**

$$\nabla^2_{Cartesian}\varphi = \partial_i\partial_i\varphi = \frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} + \frac{\partial^2\varphi}{\partial z^2}. \tag{1.5.22}$$

Physical Interpretation

The Laplacian measures how the value of φ at a point differs from its average over a small neighborhood. Consider a cube of side a centered at the origin. The average value

of φ over the cube is:

$$\bar{\varphi} = \frac{1}{a^3} \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} \varphi \, dx_1 \, dx_2 \, dx_3.$$

Expanding $\varphi(x_i) = \varphi_0 + (\partial_i \varphi)_0 x_i + \frac{1}{2}(\partial_i \partial_j \varphi)_0 x_i x_j + \cdots$ and integrating:

- Linear terms vanish (odd functions over symmetric domain)
- Cross terms $x_i x_j$ with $i \neq j$ vanish similarly
- Diagonal terms x_i^2 contribute $a^2/12$ each

The result is:

$$(\nabla^2 \varphi)_0 \approx \frac{24}{a^2}(\bar{\varphi} - \varphi_0). \tag{1.5.23}$$

Thus $\nabla^2 \varphi > 0$ means φ at the center is *less than* its neighborhood average; $\nabla^2 \varphi < 0$ means it is *greater*.

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Example 1.5.4 (Important Equations of Physics)**.**

- **Laplace's equation**: $\nabla^2 \varphi = 0$ (harmonic functions, electrostatic potential in vacuum)
- **Poisson's equation**: $\nabla^2 \varphi = \rho(x_i)$ (electrostatic potential with charge density ρ)
- **Heat equation**: $\nabla^2 \varphi = K \frac{\partial \varphi}{\partial t}$ (diffusion, heat conduction)
- **Wave equation**: $\nabla^2 \varphi = \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2}$ (electromagnetic waves, sound)
- **Helmholtz equation**: $\nabla^2 \varphi + k^2 \varphi = 0$ (time-harmonic waves)

1.5.7 Vector Laplacian and Important Identities

B&F pp. 30–32

Definition 1.5.6 (Vector Laplacian)**.** In Cartesian coordinates:

B&F Eqs. (1.7 pp. 30

$$\nabla^2_{\text{Cartesian}} \mathbf{V} \equiv \mathbf{e}_i \nabla^2 V_i. \tag{1.5.24}$$

That is, the Laplacian acts componentwise.

In curvilinear coordinates, the vector Laplacian is *defined* by:

$$\nabla^2 \mathbf{V} \equiv -\nabla \times (\nabla \times \mathbf{V}) + \nabla(\nabla \cdot \mathbf{V}). \tag{1.5.25}$$

B&F Eq. (1.79)

Theorem 1.5.11 (Curl of Curl Identity)**.** *In Cartesian coordinates:*

$$\nabla \times (\nabla \times \mathbf{V}) = -\nabla^2_{Cartesian} \mathbf{V} + \nabla(\nabla \cdot \mathbf{V}). \tag{1.5.26}$$

Proof. Using index notation in Cartesian coordinates:

$$[\nabla \times (\nabla \times \mathbf{V})]_i = \varepsilon_{ijk} \partial_j (\nabla \times \mathbf{V})_k = \varepsilon_{ijk} \partial_j \varepsilon_{klm} \partial_l V_m.$$

Using the ε-δ identity $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$:

$$\begin{aligned}
[\nabla \times (\nabla \times \mathbf{V})]_i &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \partial_j \partial_l V_m \\
&= \partial_j \partial_i V_j - \partial_j \partial_j V_i \\
&= \partial_i (\partial_j V_j) - \partial_j \partial_j V_i \\
&= [\nabla(\nabla \cdot \mathbf{V})]_i - (\nabla^2 V)_i.
\end{aligned}$$

■

1.5.8 Fundamental Identities: Curl of Gradient and Divergence of Curl

B&F pp. Eqs. (1.83)–(1.8

Theorem 1.5.12 (Curl of Gradient Vanishes)**.** *For any sufficiently smooth scalar field φ:*

B&F Eq. (1.83)

$$curl(grad\,\varphi) = \nabla \times (\nabla\varphi) = \mathbf{0}. \tag{1.5.27}$$

Proof. In Cartesian coordinates:

$$[\mathrm{curl}(\mathrm{grad}\,\varphi)]_i = \varepsilon_{ijk}\partial_j\partial_k\varphi.$$

We show this vanishes by the antisymmetry argument. Consider the sum:

$$S = \varepsilon_{ijk}\partial_j\partial_k\varphi.$$

The Levi-Civita symbol ε_{ijk} is antisymmetric under exchange of j and k: $\varepsilon_{ijk} = -\varepsilon_{ikj}$.

The mixed partial derivatives are symmetric (for smooth φ): $\partial_j\partial_k\varphi = \partial_k\partial_j\varphi$.

Thus S is the contraction of an antisymmetric tensor (ε_{ijk}) with a symmetric tensor ($\partial_j\partial_k\varphi$). Such contractions always vanish.

Explicit verification: Let $a_{ij} = -a_{ji}$ (antisymmetric) and $b_{ij} = b_{ji}$ (symmetric). Then:

$$s = a_{ij}b_{ij}.$$

Relabeling dummy indices $i \leftrightarrow j$:

$$s = a_{ji}b_{ji} = (-a_{ij})(b_{ij}) = -a_{ij}b_{ij} = -s.$$

Therefore $s = -s$, which implies $s = 0$. ■

(1.84), p. 31

Theorem 1.5.13 (Divergence of Curl Vanishes)**.** *For any sufficiently smooth vector field $\mathbf{V}$:*

$$div(curl\,\mathbf{V}) = \nabla \cdot (\nabla \times \mathbf{V}) = 0. \tag{1.5.28}$$

Proof. In Cartesian coordinates:

$$\mathrm{div}(\mathrm{curl}\,\mathbf{V}) = \partial_i(\mathrm{curl}\,\mathbf{V})_i = \partial_i\varepsilon_{ijk}\partial_j V_k = \varepsilon_{ijk}\partial_i\partial_j V_k.$$

This is again the contraction of the antisymmetric ε_{ijk} (in indices i, j) with the symmetric $\partial_i\partial_j V_k$ (symmetric in i, j by equality of mixed partials). Therefore it vanishes. ■

Physical Interpretation

These identities have important consequences:

- If $\mathbf{V} = \nabla\varphi$ (gradient field), then $\nabla \times \mathbf{V} = 0$ (the field is irrotational)
- If $\mathbf{V} = \nabla \times \mathbf{A}$ (curl of something), then $\nabla \cdot \mathbf{V} = 0$ (the field is solenoidal)

The converses are also true (under suitable conditions):

- If $\nabla \times \mathbf{V} = 0$, then $\mathbf{V} = \nabla\varphi$ for some scalar potential φ
- If $\nabla \cdot \mathbf{V} = 0$, then $\mathbf{V} = \nabla \times \mathbf{A}$ for some vector potential $\mathbf{A}$

32, Eqs. (1.85)–

. (1.85)–(1.89),

1.5.9 Application: Maxwell's Equations

The differential operators provide an elegant formulation of electromagnetism.

Theorem 1.5.14 (Maxwell's Equations)**.** *In cgs units with the displacement current:*

$$\nabla \cdot \mathbf{D} = 4\pi\rho \qquad \textit{(Coulomb's law)} \tag{1.5.29}$$

$$\nabla \cdot \mathbf{B} = 0 \qquad \textit{(no magnetic monopoles)} \tag{1.5.30}$$

$$\nabla \times \mathbf{E} = -\frac{1}{c}\frac{\partial \mathbf{B}}{\partial t} \qquad \textit{(Faraday's law)} \tag{1.5.31}$$

$$\nabla \times \mathbf{H} = \frac{4\pi}{c}\mathbf{J} + \frac{1}{c}\frac{\partial \mathbf{D}}{\partial t} \qquad \textit{(Ampère-Maxwell law)} \tag{1.5.32}$$

Physical Interpretation

The identity $\nabla \cdot (\nabla \times \mathbf{V}) = 0$ ensures consistency:

- From (1.5.30): $\mathbf{B} = \nabla \times \mathbf{A}$ for some vector potential $\mathbf{A}$
- Taking divergence of (1.5.32) and using $\nabla \cdot (\nabla \times \mathbf{H}) = 0$ gives the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

Maxwell's addition of the displacement current term $\frac{1}{c}\frac{\partial \mathbf{D}}{\partial t}$ was precisely to ensure this consistency, and it led to the prediction of electromagnetic waves.

1.5.10 Key Results Summary

- **Gradient**: $(\operatorname{grad}\varphi)_i = \frac{1}{h_i}\partial_i\varphi$; points in direction of maximum rate of change
- **Divergence**: $\operatorname{div}\mathbf{V} = \frac{1}{h_1h_2h_3}\sum_i \partial_i(h_jh_kV_i)$; measures source strength per unit volume
- **Curl**: $(\operatorname{curl}\mathbf{V})_i = \frac{1}{h_jh_k}[\partial_j(h_kV_k) - \partial_k(h_jV_j)]$; measures circulation per unit area
- **Laplacian**: $\nabla^2\varphi = \operatorname{div}(\operatorname{grad}\varphi)$; measures deviation from local average
- **Gauss's theorem**: $\int_\tau \operatorname{div}\mathbf{V}\,d\tau = \oint_\sigma \mathbf{V}\cdot d\boldsymbol{\sigma}$
- **Stokes's theorem**: $\int_\sigma \operatorname{curl}\mathbf{V}\cdot d\boldsymbol{\sigma} = \oint_\lambda \mathbf{V}\cdot d\boldsymbol{\lambda}$
- **Fundamental identities**: $\nabla \times (\nabla\varphi) = \mathbf{0}$ and $\nabla \cdot (\nabla \times \mathbf{V}) = 0$
- **Curl of curl**: $\nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - \nabla^2\mathbf{V}$